Continuous integration is a foundational practice of continuous delivery. When adopted successfully, developers work in small steps and commit to the mainline (the main branch) multiple times a day. After each code commit, the commit stage of the deployment pipeline takes care of building and testing the application with the new changes.

This stage should be fast, because a developer will wait until it completes successfully before moving on to their next task. That's a critical point. If the commit stage fails, the developer responsible for it should immediately deliver a fix or revert their changes so as not to leave the mainline in a broken state and prevent all other developers from integrating their code.

Let's start designing a deployment pipeline for a cloud native application like Catalog Service. For now, we'll focus on the first few steps in the commit stage (figure 3.8).

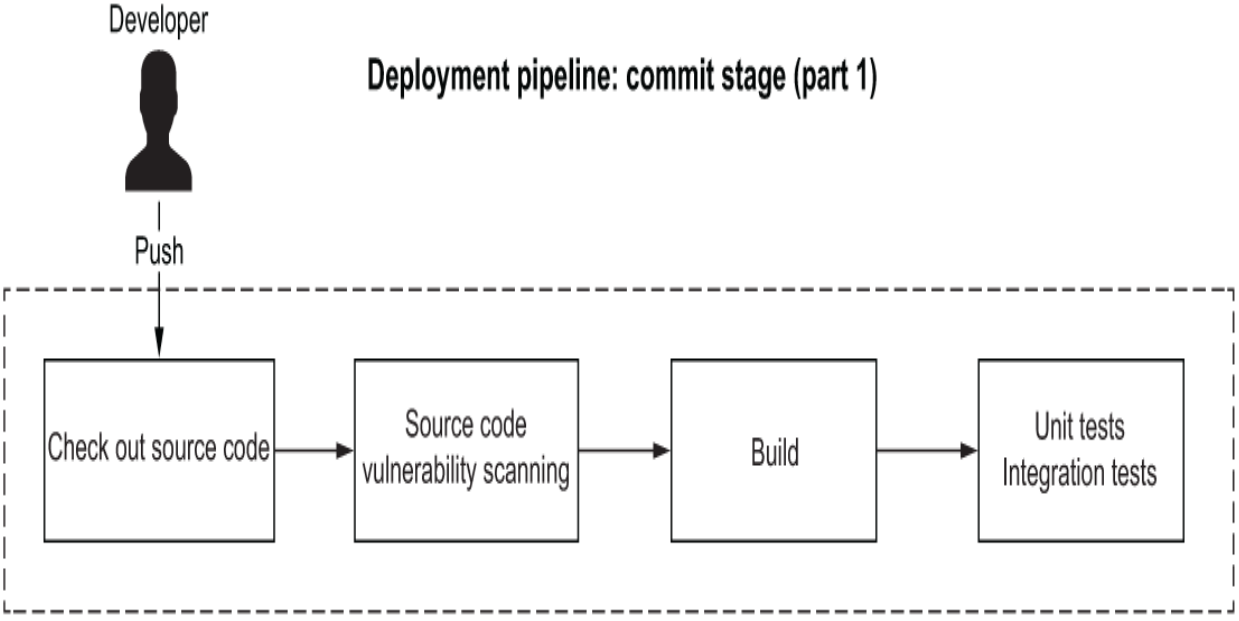


Figure 3.8 The first part of the commit stage in a deployment pipeline